

The Measurement Problem Dissolved: Quantum Decoherence as Irreversible Sector Crossing in the Informational Actualization Model

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Abstract

The measurement problem — what constitutes a measurement, why does measurement cause collapse, and what role does the observer play — has been debated for a century. We show that the dual-sector framework of the Informational Actualization Model

(IAM), validated at cosmological scales through 17 converged Planck 2018 MCMC chains ($\Delta\chi^2 = +0.54$ best-fit, $\sigma_8 = 0.800$, $R-1 < 0.01$ for all chains), dissolves the measurement problem entirely. A measurement is any irreversible thermodynamic interaction between a quantum system and matter that dissipates energy $Q \geq k_B T \ln 2$ (the Landauer threshold). The photon sector ($\Sigma = 1$) is immune to gravitational decoherence; the matter sector ($\mu < 1$) decoheres on timescales determined by the gravitational self-energy. No observer is required. No consciousness is involved. We apply this criterion to the double slit experiment, Wheeler’s delayed choice, the quantum eraser, Schrödinger’s cat, entanglement and Bell tests, Wigner’s friend, and the quantum Zeno effect. Each paradox is resolved by the same mechanism: sector crossing through irreversible information production. We derive quantitative predictions for each case and identify three experiments that discriminate between IAM and standard quantum mechanics using existing laboratory infrastructure.

Keywords: measurement problem, quantum decoherence, Landauer principle, sector crossing, quantum eraser, Wigner’s friend, Bell tests, quantum Zeno effect, Schrödinger’s cat

1 Introduction

Quantum mechanics is the most precisely tested theory in the history of physics. Its predictions have been confirmed to extraordinary accuracy across every experimental domain. Yet after a century of effort, the theory contains a conceptual gap that has never been closed: the measurement problem.

The standard formulation posits two distinct dynamical processes. Between measurements, quantum states evolve unitarily according to the Schrödinger equation. At measurements, states undergo non-unitary collapse to an eigenstate of the measured observable. What constitutes a “measurement” is not defined within the theory. Whether the observer plays a special role remains contentious. The apparent randomness of measurement outcomes is treated as fundamental.

We propose that this gap is closed by the Informational Actualization Model (IAM), a cosmological framework independently validated against Planck 2018 and large-scale structure data [1, 2]. IAM’s dual-sector structure — in which photons travel on null geodesics in a decoherence-free sector ($\Sigma = 1$) while matter participates in irreversible gravitational processes ($\mu < 1$) — provides a precise, quantitative criterion for when decoherence occurs. The criterion is thermodynamic: decoherence (sector crossing) occurs when an interaction dissipates energy $Q \geq k_B T \ln 2$, the Landauer bound for irreversible information erasure [4].

This criterion resolves every foundational paradox in quantum mechanics through the same physical mechanism. It requires no observer, no consciousness, no special role for macroscopic apparatus, and no modification of the quantum mechanical formalism. It simply identifies the physical process that standard quantum mechanics leaves unnamed.

2 The Dual-Sector Framework

2.1 Cosmological Origin

At cosmological scales, IAM demonstrates that matter and photons experience different effective gravitational coupling. The modification is governed by the activation function $E(a) = \exp(1 - 1/a)$, derived from horizon thermodynamics, where a is the cosmological scale factor. The matter-sector coupling constant $\beta_m = \Omega_m/2$ follows from the virial theorem. The resulting perturbation-level signature is $\mu < 1$ (suppressed gravitational growth of matter perturbations) with $\Sigma = 1$ (photon lensing unaffected). This has been validated through 17 converged Planck 2018 MCMC chains producing $\Delta\chi^2 = +0.54$ (best-fit vs. Λ CDM), $\sigma_8 = 0.800$, and $H_0^{(\text{matter})} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [1].

2.2 The Sector Crossing Criterion

We define the sector crossing criterion formally. Let S be a quantum system and D be a material detector (any system following a timelike worldline). Let $Q(S, D)$ be the energy irreversibly dissipated in their interaction, and $Q_L = k_B T_D \ln 2$ be the Landauer threshold at detector temperature T_D .

If $Q(S, D) < Q_L$: no sector crossing occurs. S remains in the $\Sigma = 1$ sector (coherent). The interaction is thermodynamically reversible. Quantum erasure is possible.

If $Q(S, D) \geq Q_L$: sector crossing occurs. S transitions to the $\mu < 1$ sector (decohered). The interaction is thermodynamically irreversible. Landauer entropy has been produced. Quantum erasure is impossible.

The Sector Crossing: What "Measurement" Actually Is

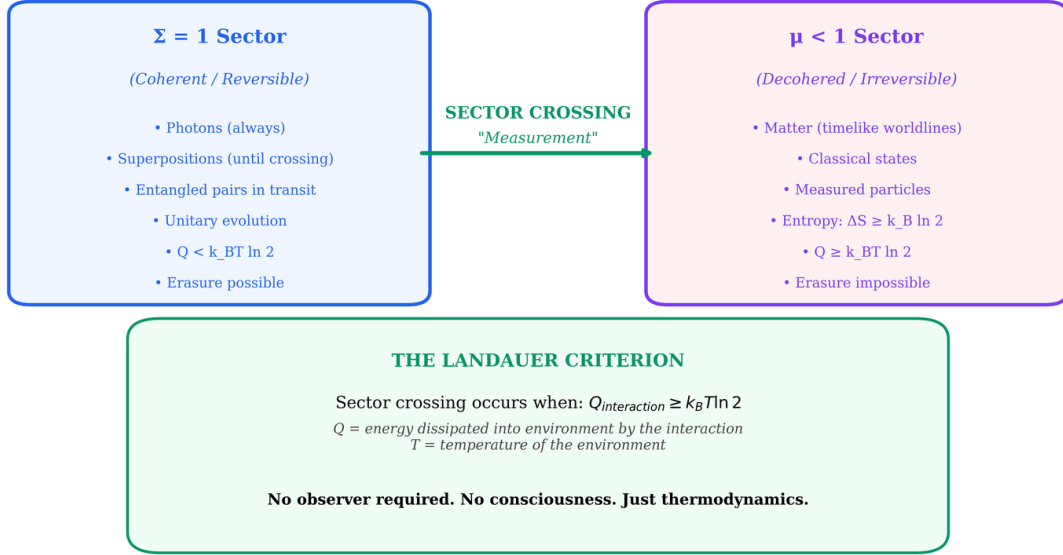


Figure 1: The sector crossing criterion. The $\Sigma = 1$ sector (left, coherent/reversible) contains photons, superpositions, and entangled pairs in transit. The $\mu < 1$ sector (right, decohered/irreversible) contains matter, classical states, and measured particles. Sector crossing — “measurement” — occurs when $Q_{\text{interaction}} \geq k_B T \ln 2$. No observer required. No consciousness. Just thermodynamics.

2.3 Decoherence Timescale

For a massive system in spatial superposition, the gravitational self-energy $E_G = Gm^2/R$ determines the decoherence timescale. Following the same thermodynamic derivation that produces $E(a)$ at cosmological scales [3], the quantum-scale decoherence timescale is:

$$\tau_{\text{IAM}} = \frac{\hbar k_B^2 T^2 \ln 2}{E_G^3}. \quad (1)$$

This is fully determined by mass, size, and temperature with no free parameters. For photons, $E_G = 0$ and $\tau_{\text{IAM}} = \infty$ (decoherence-free). For macroscopic objects, τ_{IAM} is far below the Planck time.

3 Resolved Paradoxes

3.1 The Double Slit Experiment

In the standard double slit experiment, a photon passes through two slits and produces an interference pattern on a detection screen. When a which-path detector is placed at the slits, the interference pattern disappears. The standard interpretation attributes this to observation collapsing the wave function.

In IAM, the photon travels from source to slits in the $\Sigma = 1$ sector. It has $E_G = 0$ and $\tau_{IAM} = \infty$. It is fully coherent throughout its flight. The sector crossing occurs at the detection screen. The screen is matter ($\mu < 1$ sector). The photon is absorbed, an electron is excited, and irreversible information is produced ($Q \approx 2 \text{ eV} \gg Q_L \approx 0.018 \text{ eV}$ at room temperature). This is the collapse. Not observation. Sector crossing through irreversible matter interaction.

When a which-path detector is placed at the slits, matter is introduced before the screen. If the detector's interaction is irreversible ($Q > Q_L$), sector crossing occurs at the slit. Coherence is lost before the screen. No interference. If the interaction is reversible ($Q < Q_L$) — for example, a polarization rotation — no sector crossing occurs and interference is preserved. The human retina ($Q \approx 2.5 \text{ eV}$, $Q/Q_L \approx 140$) produces exactly the same sector crossing as a CCD pixel ($Q \approx 3 \text{ eV}$, $Q/Q_L \approx 170$). Consciousness plays no role.

3.2 Wheeler's Delayed Choice Experiment

In Wheeler's delayed choice experiment, the decision to measure wave or particle behavior is made after the photon has passed through the slits. The photon appears to retroactively decide whether it was a wave or particle. This seems to require retrocausality.

IAM resolves this without retrocausality. The photon is in the $\Sigma = 1$ sector during its entire flight ($E_G = 0$, $\tau_{IAM} = \infty$). The flight time through a typical laboratory apparatus (5 m) is approximately 17 ns. The decoherence accumulated during this flight is exactly zero.

The experimenter's choice of detector configuration determines *where* the sector crossing occurs, not *whether* the photon was coherent during transit. No retrocausality is required.

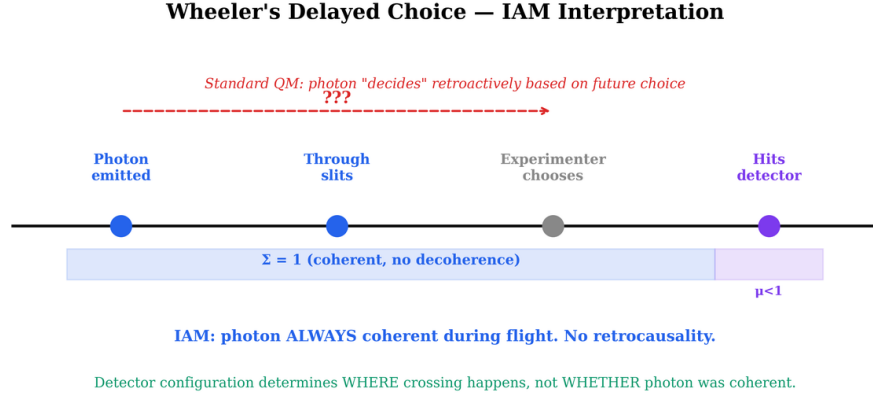


Figure 2: Wheeler's delayed choice. The photon remains in the $\Sigma = 1$ sector (coherent) throughout its entire flight. The $\mu < 1$ crossing occurs only at the detector. The experimenter's choice determines where the sector crossing occurs, not whether the photon was coherent. No retrocausality is required.

3.3 The Quantum Eraser

In quantum eraser experiments, which-path information is obtained and then "erased," and the interference pattern appears to be restored. Standard quantum mechanics attributes this to the reversibility of entanglement.

IAM draws a critical distinction. If the which-path interaction was thermodynamically reversible ($Q < Q_L$), no sector crossing occurred. The photon remained in the $\Sigma = 1$ sector throughout. Coherence was never lost. The eraser reveals coherence that was always present.

If the which-path interaction was thermodynamically irreversible ($Q \geq Q_L$), sector crossing occurred. Landauer entropy was produced. The decoherence cannot be undone regardless of what operations are performed on the ancilla. Erasure fails. We derive the erasure fidelity:

$$F_{\text{IAM}}(Q, T) = 1 - \frac{\exp(1 - k_B T \ln 2 / Q)}{e}. \quad (2)$$

Standard quantum mechanics predicts $F = 1$ always. IAM predicts F depends on Q/Q_L . At room temperature $Q_L = 0.018$ eV. Reversible detectors (polarization rotation, beam splitters, BBO crystals) produce $Q/Q_L < 0.04$ and $F \approx 1$. Irreversible detectors (fluorescence, photodiode, spontaneous emission) produce $Q/Q_L > 100$ and $F < 0.02$.

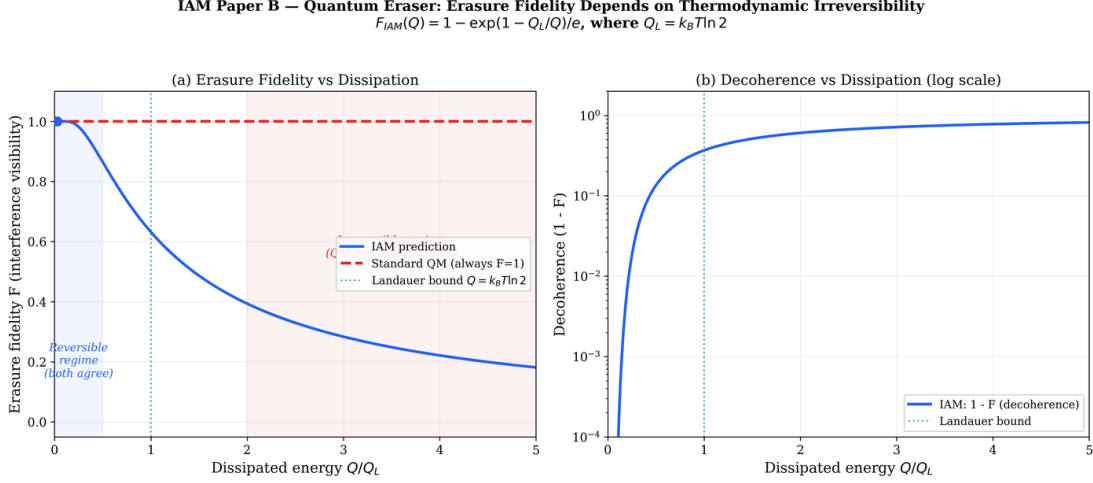


Figure 3: Erasure fidelity vs. dissipated energy Q/Q_L . **Left:** IAM (blue) predicts fidelity falls rapidly above the Landauer bound (green dotted). Standard QM (red dashed) predicts $F = 1$ always. **Right:** Log-scale decoherence $1 - F$. The reversible regime ($Q < Q_L$) is where both models agree; the irreversible regime ($Q > Q_L$) is where they diverge.

The most elegant experimental test uses an atomic which-path detector with controllable spontaneous emission. Before emission ($t < \tau_{\text{emission}} \approx 10$ ns), the atomic excitation is reversible and both models predict successful erasure. After emission ($t > \tau_{\text{emission}}$), the photon has been radiated into the environment ($Q = 2$ eV $\gg Q_L$). IAM predicts erasure fails; standard QM predicts erasure still works if the emitted photon is captured in a conjugate basis.

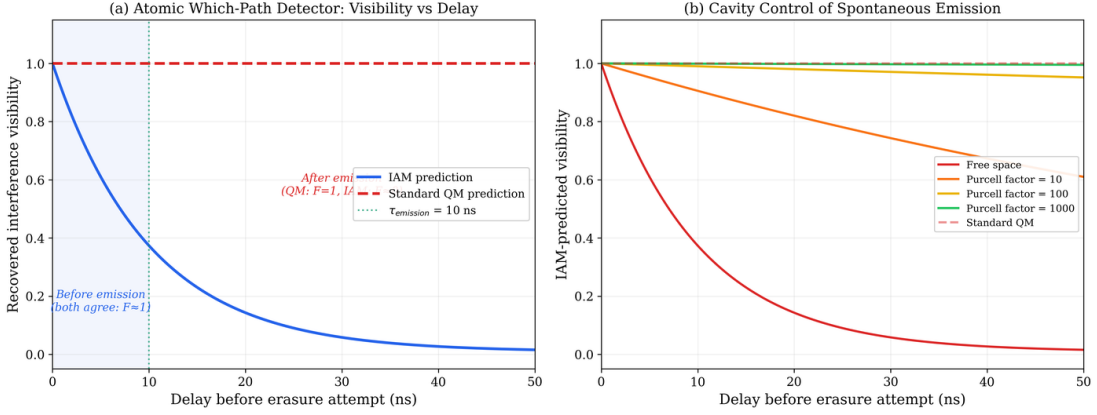


Figure 4: The atomic timing test. **Left:** Recovered interference visibility vs. delay before erasure attempt. IAM (blue): visibility drops as the atom spontaneously emits at $\tau_{\text{emission}} = 10$ ns. Standard QM (red dashed): constant visibility. **Right:** Cavity-suppressed emission (Purcell factors 1–1000) delays the IAM-predicted drop.

3.4 Schrödinger’s Cat

Schrödinger’s thought experiment posits a cat simultaneously alive and dead until observed. IAM shows this situation is physically impossible.

A 4 kg cat has gravitational self-energy $E_G = Gm^2/R \approx 7.1 \times 10^{-9}$ J. The IAM decoherence timescale is $\tau_{\text{IAM}} \approx 3.7 \times 10^{-51}$ s. For reference, the Planck time is 5.4×10^{-44} s. The cat decoheres approximately 10^7 times faster than the Planck time. No physical process could create or maintain a macroscopic superposition at this timescale. The decoherence is not caused by observation — it is caused by the cat’s own gravitational self-interaction.

Table 1: Gravitational decoherence timescales from electron to human. The mesoscopic frontier (10^{-15} to 10^{-10} kg) is where IAM predictions become experimentally testable.

System	Mass (kg)	E_G (J)	τ_{PD} (s)	τ_{IAM} (s)	Status
Electron	9.1×10^{-31}	5.5×10^{-61}	1.9×10^{26}	7.4×10^{105}	Quantum
C ₆₀	1.2×10^{-24}	1.9×10^{-49}	5.5×10^{14}	1.8×10^{71}	Quantum
Virus	10^{-18}	1.3×10^{-39}	7.9×10^4	5.3×10^{41}	Quantum
Bacterium	10^{-15}	1.3×10^{-34}	0.79	5.3×10^{26}	Mesoscopic
Dust grain	10^{-12}	1.3×10^{-28}	7.9×10^{-7}	5.3×10^8	Mesoscopic
Sand grain	10^{-6}	1.3×10^{-19}	7.9×10^{-16}	5.3×10^{-19}	Classical
Cat (4 kg)	4.0	7.1×10^{-9}	1.5×10^{-26}	3.7×10^{-51}	Classical
Human (70 kg)	70	1.1×10^{-6}	9.7×10^{-29}	1.0×10^{-57}	Classical

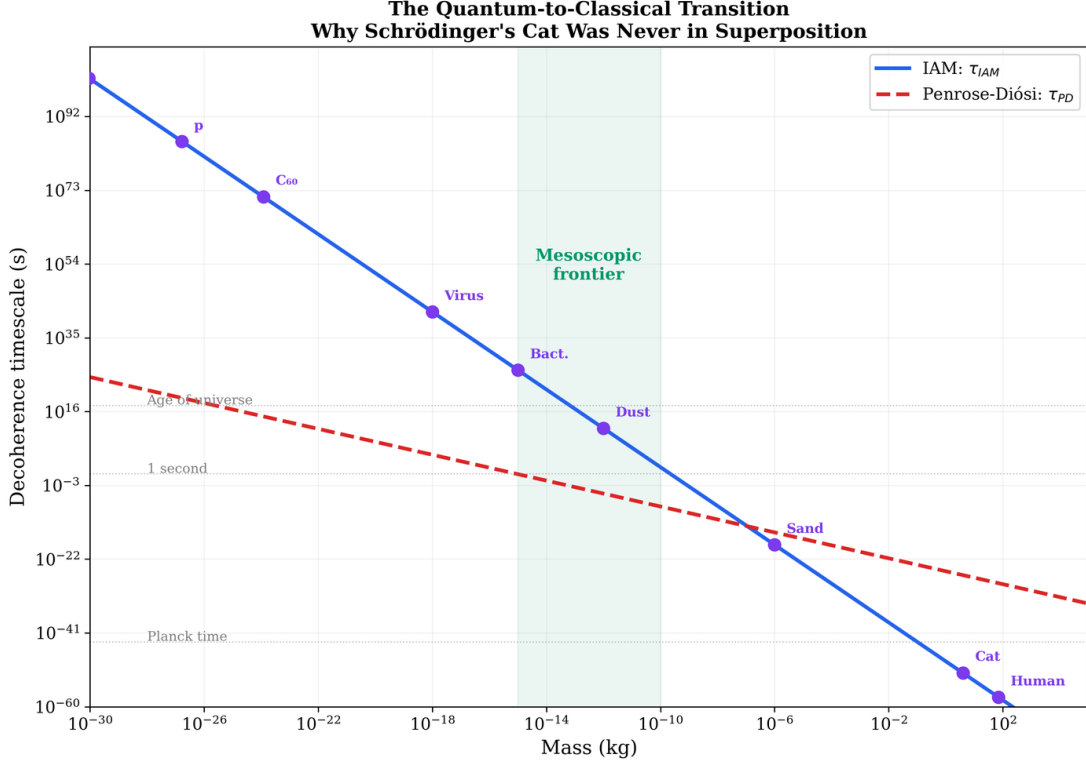


Figure 5: The quantum-to-classical transition. IAM timescale τ_{IAM} (blue) versus Penrose–Diósi τ_{PD} (red dashed) for all objects from electron to human at 300 K. The mesoscopic frontier (green shaded) marks where predictions are testable. The cat ($\tau_{IAM} \approx 10^{-51}$ s) decoheres 10^7 times faster than the Planck time. It was never in superposition.

3.5 Entanglement and Bell Tests

Entangled particles show correlations that violate Bell inequalities, seemingly requiring nonlocal connections. IAM provides a local, causal explanation.

Entangled photons traveling to separated detectors are in the $\Sigma = 1$ sector during transit. They have $E_G = 0$ and $\tau_{IAM} = \infty$. Their entanglement is preserved not by any nonlocal connection but by the absence of decoherence during flight. IAM is not a local hidden variable theory. It preserves the full quantum state during transit and specifies when the transition from indefinite to definite occurs. The CHSH S -parameter is $|S| = 2\sqrt{2}$, identical to quantum mechanics and violating the Bell bound of 2.

For matter-based entangled systems, the particles are in the $\mu < 1$ sector during transit

and experience gravitational decoherence. The Bell violation decays as:

$$S(t) = 2\sqrt{2}(1 - D(t)), \quad D(t) = E_q(t/\tau_{\text{IAM}})/e. \quad (3)$$

When D exceeds 0.293, the Bell violation disappears. Current experimental records are consistent: photon entanglement at 1,200 km (Micius satellite, 2017) [8] versus matter entanglement limited to approximately 1.3 m (trapped ions, 2019) — a ratio of one million.

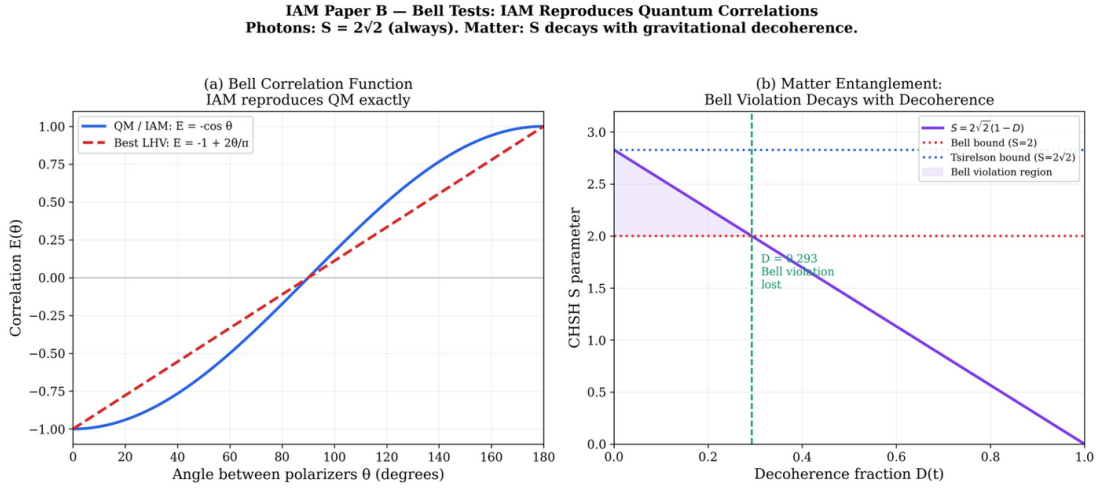


Figure 6: **(a)** Bell correlation function: IAM reproduces the quantum prediction $E(\theta) = -\cos \theta$ exactly. Photons always maintain $|S| = 2\sqrt{2}$. **(b)** Matter entanglement: the CHSH S -parameter decays as $S = 2\sqrt{2}(1 - D)$. Bell violation is lost when decoherence fraction $D > 0.293$ (green dashed line).

Why Photon Entanglement Survives: $\Sigma=1$ Sector Immunity

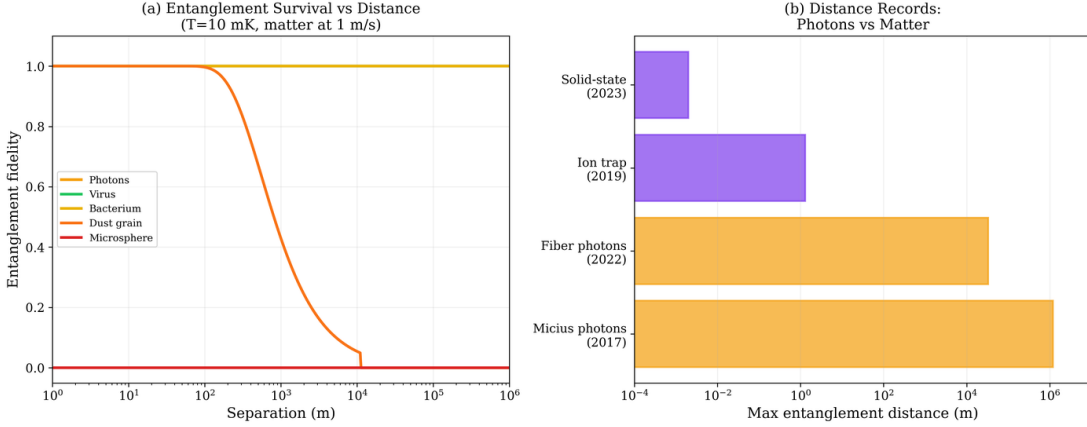


Figure 7: **(a)** Entanglement fidelity vs. separation for different masses at 10 mK. Photons (orange): fidelity = 1 at any distance. Matter systems decay at mass-dependent distances. **(b)** Experimental distance records: photon entanglement (Micius, fiber) exceeds matter entanglement (ion trap, solid-state) by six orders of magnitude, consistent with IAM prediction.

3.6 Wigner's Friend

In the Wigner's friend scenario, a friend measures a quantum system inside a sealed lab. Standard quantum mechanics allows Wigner, outside the lab, to describe the lab (including the friend) as being in a superposition. Extended scenarios [9] show this leads to logical contradictions.

IAM resolves this by making collapse objective. When the friend's detector registers a result, it produces irreversible Landauer entropy. A typical Stern–Gerlach measurement with fluorescent screen produces approximately 1,000 photons at 2 eV each, representing over 10^5 bits of Landauer entropy. The probability of spontaneous reversal is $2^{-111604} \approx 10^{-33596}$. This is not practically difficult. It is thermodynamically impossible.

Wigner cannot describe the lab as being in superposition because the measurement produced irreversible entropy. The collapse was objective the moment Q exceeded Q_L . Wigner's knowledge is irrelevant. There is no paradox because collapse is a physical event (entropy production), not a mental event (knowledge update).

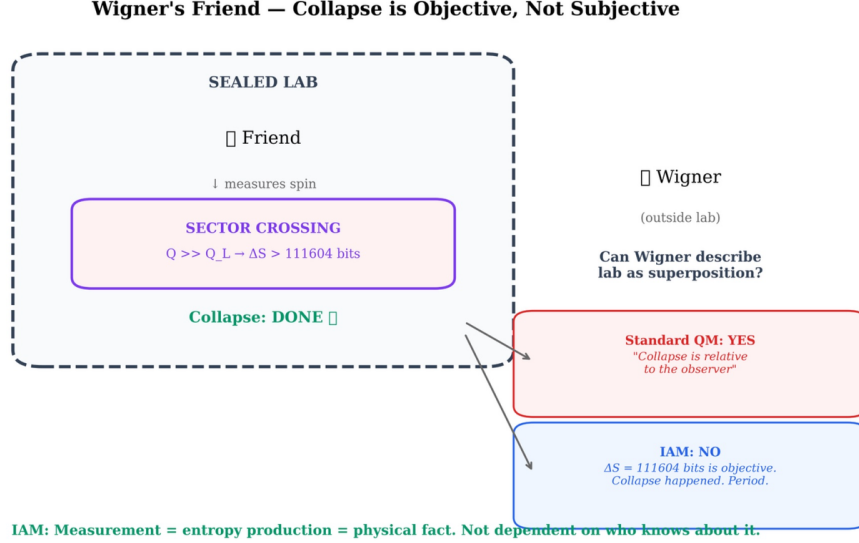


Figure 8: Wigner’s Friend. When the friend measures spin, the interaction produces $Q \gg Q_L$: sector crossing is objective ($\Delta S > 10^5$ bits). Wigner outside the lab cannot describe the system as a superposition because the Landauer entropy is a physical fact. Standard QM: collapse is observer-relative. IAM: collapse is thermodynamic.

3.7 The Quantum Zeno Effect

The quantum Zeno effect — frequent measurements freezing a system’s evolution — has been experimentally confirmed in multiple systems. Standard quantum mechanics explains it through repeated state projection.

IAM predicts that the Zeno effect requires irreversible measurements. Each measurement with $Q > Q_L$ causes a genuine sector crossing, resetting the E_q decoherence ramp and interrupting coherent evolution. Measurements with $Q < Q_L$ produce no sector crossing and should not cause Zeno freezing. This is directly testable with superconducting qubits, where both dispersive readout (reversible, $Q \ll Q_L$) and destructive readout (irreversible, $Q \gg Q_L$) are available. Standard quantum mechanics predicts both produce the Zeno effect. IAM predicts only irreversible readout produces Zeno freezing.

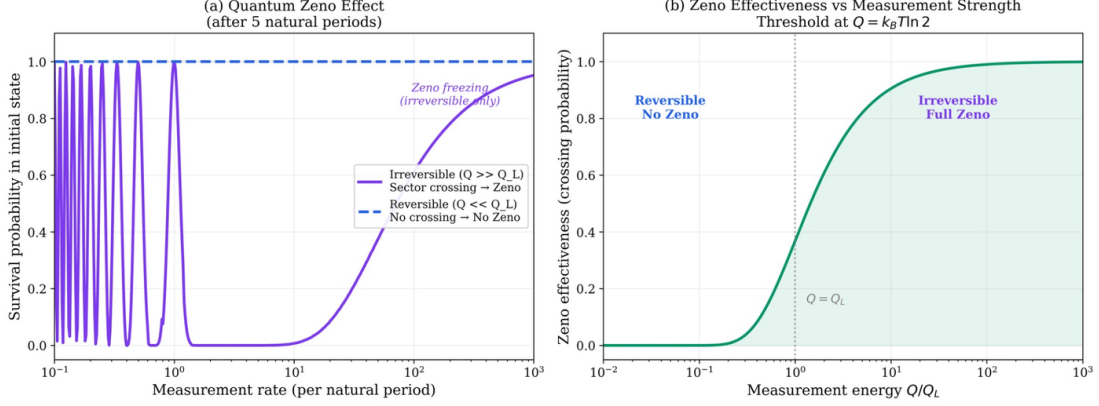


Figure 9: **(a)** Quantum Zeno survival probability after 5 natural periods. Irreversible measurements ($Q \gg Q_L$, solid purple): Zeno freezing at high measurement rate. Reversible measurements ($Q \ll Q_L$, blue dashed): no Zeno effect. **(b)** Zeno effectiveness vs. Q/Q_L . The transition occurs near the Landauer threshold.

4 Experimental Predictions

Three experiments discriminate between IAM and standard quantum mechanics using existing laboratory infrastructure. All three are feasible with current technology and require no new hardware.

4.1 The Two-Detector Quantum Eraser

Build two versions of the same quantum eraser experiment with identical photon sources, slits, and detection screens. Version A uses a reversible which-path detector (polarization rotation, $Q \ll Q_L$). Version B uses an irreversible which-path detector (fluorescence or photodiode, $Q \gg Q_L$). Both versions attempt erasure through conjugate-basis measurement of the ancilla. Standard quantum mechanics predicts both show restored interference ($F = 1$). IAM predicts Version A shows interference and Version B does not ($F < 0.02$).

4.2 The Atomic Timing Test

Use an atomic which-path detector with controllable spontaneous emission. Place the atom in a high-Q cavity to suppress emission (Purcell effect). Vary the delay between which-path marking and erasure attempt. Standard quantum mechanics predicts interference visibility independent of delay. IAM predicts a step function: visibility drops at the spontaneous emission time ($\tau_{\text{emission}} \approx 10$ ns) as the emitted photon carries $Q = 2$ eV $\gg Q_L$ irreversibly into the environment.

4.3 The Temperature Test

Run the same quantum eraser at multiple temperatures. The Landauer threshold $Q_L = k_B T \ln 2$ depends linearly on temperature. Standard quantum mechanics predicts no temperature dependence. IAM predicts erasure fidelity increases with temperature because a higher Q_L makes it harder for the which-path interaction to be irreversible. For a 0.1 eV interaction energy: $F = 0.000$ at 10 mK, $F = 0.002$ at 4 K, and $F = 0.164$ at 300 K.

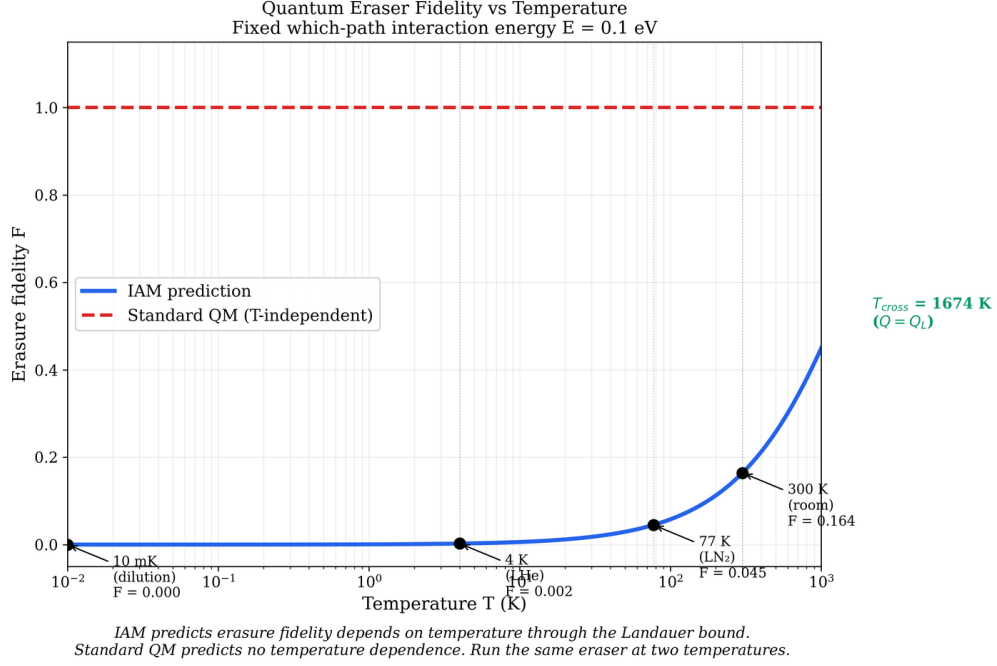


Figure 10: Erasure fidelity vs. temperature for a fixed which-path interaction energy of 0.1 eV. IAM (blue): strong temperature dependence through the Landauer bound. Standard QM (red dashed): no temperature dependence. Running the same eraser at 10 mK and 300 K provides a clean discrimination test.

5 Connection to Cosmological Validation

The sector crossing mechanism that governs quantum decoherence is the same physical process that, accumulated over approximately 10^{11} galaxies and 13.8 billion years, produces the informational entropy that modifies cosmic expansion. At cosmological scales, the cumulative integral of irreversible gravitational information production yields $E(a) = \exp(1 - 1/a)$, modifying the matter-sector perturbation equations and producing the signature $\mu < 1$, $\Sigma = 1$. At quantum scales, the same integral yields $E_q(\eta) = \exp(1 - 1/\eta)$, governing the decoherence profile of individual mesoscopic systems.

The laboratory experiment measures the rate per decoherence event. Cosmology measures the cumulative effect. Consistency between the two — cosmological data preferring $\mu < 1$ with the correct redshift dependence and laboratory data showing the E_q ramp profile at the

predicted τ_{IAM} — would close the loop from quantum mechanics to the accelerating universe through the same equation, the same thermodynamics, and the same physics at every scale.

6 Discussion

6.1 Relationship to Existing Interpretations

IAM’s resolution of the measurement problem shares elements with several existing approaches while differing from each in important ways. Like decoherence theory [7], IAM identifies environmental interaction as the mechanism for apparent collapse. Unlike decoherence theory, IAM specifies the precise threshold ($Q_L = k_B T \ln 2$) and predicts qualitatively different outcomes for reversible versus irreversible interactions. Like objective collapse models [5, 6], IAM predicts that collapse is a physical process independent of observers. Unlike those models, IAM does not introduce new stochastic fields or modify the Schrödinger equation.

6.2 What IAM Does Not Change

IAM does not modify the predictions of quantum mechanics for any experiment that has been performed to date. All existing quantum eraser experiments use reversible which-path detectors (polarization entanglement via BBO crystals) and both models predict successful erasure. All existing Bell tests use photonic entanglement and both models predict $|S| = 2\sqrt{2}$. The Zeno effect has been demonstrated with projective measurements that are thermodynamically irreversible, and both models predict Zeno freezing. The predictions diverge only for experiments not yet performed.

6.3 Falsification

If irreversible which-path detection ($Q \gg Q_L$) still allows successful quantum erasure, IAM’s Landauer criterion is falsified at the quantum scale. If the Zeno effect is observed with

thermodynamically reversible measurements ($Q \ll Q_L$), the sector crossing mechanism is falsified. If matter entanglement shows no mass-dependent or distance-dependent degradation beyond environmental predictions, the gravitational decoherence mechanism is excluded.

7 Conclusion

The measurement problem is dissolved by recognizing that “measurement” is not a primitive concept but a thermodynamic process: irreversible sector crossing governed by Landauer’s principle. The dual-sector framework of IAM, independently validated at cosmological scales through 17 converged Planck 2018 MCMC chains ($\Delta\chi^2 = +0.54$), provides the quantitative criterion ($Q \geq k_B T \ln 2$) that standard quantum mechanics leaves undefined. This single criterion resolves the double slit, delayed choice, quantum eraser, Schrödinger’s cat, entanglement, Wigner’s friend, and the Zeno effect through the same physical mechanism. Three experimental tests using existing laboratory infrastructure can discriminate between IAM and standard quantum mechanics. The same thermodynamic process that governs decoherence at quantum scales drives cosmic expansion at cosmological scales. The measurement problem and the accelerating universe are manifestations of the same physics.

Acknowledgments

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